Started on	2023-10-15 19:12
State	Finished
Completed on	2023-10-16 03:37
Time taken	8 hours 25 mins
Grade	**4.00** out of 5.00 (**80**%)

Question **1**
Complete
Mark 0.00 out of 1.00

Choose a correct substitution, which allows to move from Cauchy-Euler equation to the equation with constant coefficients.

Select one:

- $t \to \ln t$
- $t \to e^t$
- $t \to t^2$
- none of the above

Your answer is incorrect.

Question **2**
Complete
Mark 1.00 out of 1.00

When we solve the equation $y'' + p(t)y' + q(t) = 0$ using reduction of order we must

Select one:

- find y_1 and y_2 by guessing.
- find at least y_1 by guessing and there is a formula for y_2.
- there are formulas for both y_1 and y_2, so we do not have to guess anything.
- none of the above

Your answer is correct.

Question 3
Complete
Mark 1.00 out of 1.00

The motion of free undamped mechanical vibrations is described by the function $y(t) = A\sin(\omega t + \varphi).$

Match the letters with corresponding names.

ω	angular frequency
A	amplitude
φ	phase angle

Your answer is correct.

Question 4
Complete
Mark 1.00 out of 1.00

Choose correct name of the motion given by the differential equation $10y'' + 3y' + 49y = 0.$

Select one:
- undamped
- underdamped (selected)
- critically damped
- overdamped
- forced

Your answer is correct.

Question 5
Complete
Mark 1.00 out of 1.00

For which γ the system described by $y'' + 4y = 5\cos\gamma t$ will be at resonance?

Select one:
- 1
- 2 (selected)
- 3
- 4

Your answer is correct.